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Intravaginal Longitudinal Physiology Monitoring Wearable Vibration, Haptic, and Internal Stimulus Integration

Field

This disclosure relates to intravaginal physiological monitoring devices and, more particularly, to the integration of vibration, haptic, and internal mechanical stimulus systems within a longitudinally worn intravaginal sensor platform.

Overview

In certain embodiments, the intravaginal longitudinal physiology monitoring device further incorporates one or more vibration, haptic, or active mechanical actuation elements. These elements may operate independently or in coordination with internal physiological sensing subsystems to provide controlled internal tactile stimulus, arousal activation, neuromodulation, pelvic engagement, biofeedback, or user signaling.
The device is configured for extended intravaginal wear and is designed to maintain comfort, positional stability, and measurement integrity while delivering optional internal stimulus.

Actuator Technologies

The vibration or haptic subsystem may include one or more of the following actuator types, individually or in combination:

- Miniature Linear Resonant Actuators (LRA)
- Miniature Eccentric Rotating Mass (ERM) vibration motors
- Piezoelectric bending, surface, or stack actuators
- Circumferential vibration rings or bands
- Distributed micro-actuator arrays embedded along the device body

Actuator selection may be based on size constraints, power consumption, frequency response, and compatibility with sealed, fluid-resistant device construction.

Actuator Distribution and Placement

One or more actuators may be positioned within the intravaginal device, including but not limited to:

- Longitudinal placement along the insertable body
- Circumferential placement to generate radial or uniform stimulus
- Zoned placement aligned with internal pressure, impedance, hydration, or activity sensors
- Segment-based actuation enabling spatially differentiated stimulus patterns

Actuators may be embedded within compliant materials, encapsulated within elastomeric layers, or mechanically isolated to preserve sensor accuracy.

Stimulus Characteristics

The vibration or haptic output may be modulated across one or more stimulus dimensions, including:

- Frequency and resonance tuning
- Amplitude or displacement
- Pulsed, continuous, ramped, or patterned waveforms
- Spatial sequencing or phase-shifted actuation between multiple zones

Stimulus profiles may be preconfigured, user-adjustable, or dynamically controlled.

Functional Roles

The vibration or haptic subsystem may serve one or more functional purposes, including:

- Arousal activation or sensory engagement
- Pelvic floor neuromuscular activation or relaxation
- Biofeedback correlated with pelvic activity, pressure distribution, impedance, or hydration metrics
- User alerts, confirmations, or session-state signaling

Closed-Loop and Sensor-Coupled Operation

In certain embodiments, vibration or haptic output is controlled in a closed-loop manner using real-time sensor input, including but not limited to pressure sensors, bio-impedance sensors, capacitive sensing, temperature sensors, or inertial measurements. Sensor fusion techniques may be used to adjust stimulus parameters dynamically while compensating for motion-induced artifacts.

Safety, Comfort, and Control

The device may incorporate safety mechanisms such as maximum intensity limits, duty-cycle constraints, thermal monitoring, user override controls, automatic deactivation conditions, or context-aware inhibition to ensure safe and comfortable operation during extended intravaginal wear.